TradifyERP

Complete System Overview & Module Linkages

System Architecture Document

System Version:	TradifyERP v1.0
Total Modules:	12 Modules
Database:	PostgreSQL / SQLite
Framework:	Django 5.2.8
Python Version:	3.14
Generated:	November 22, 2025

Tradify Global
Enterprise Resource Planning System

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1. Executive Summary

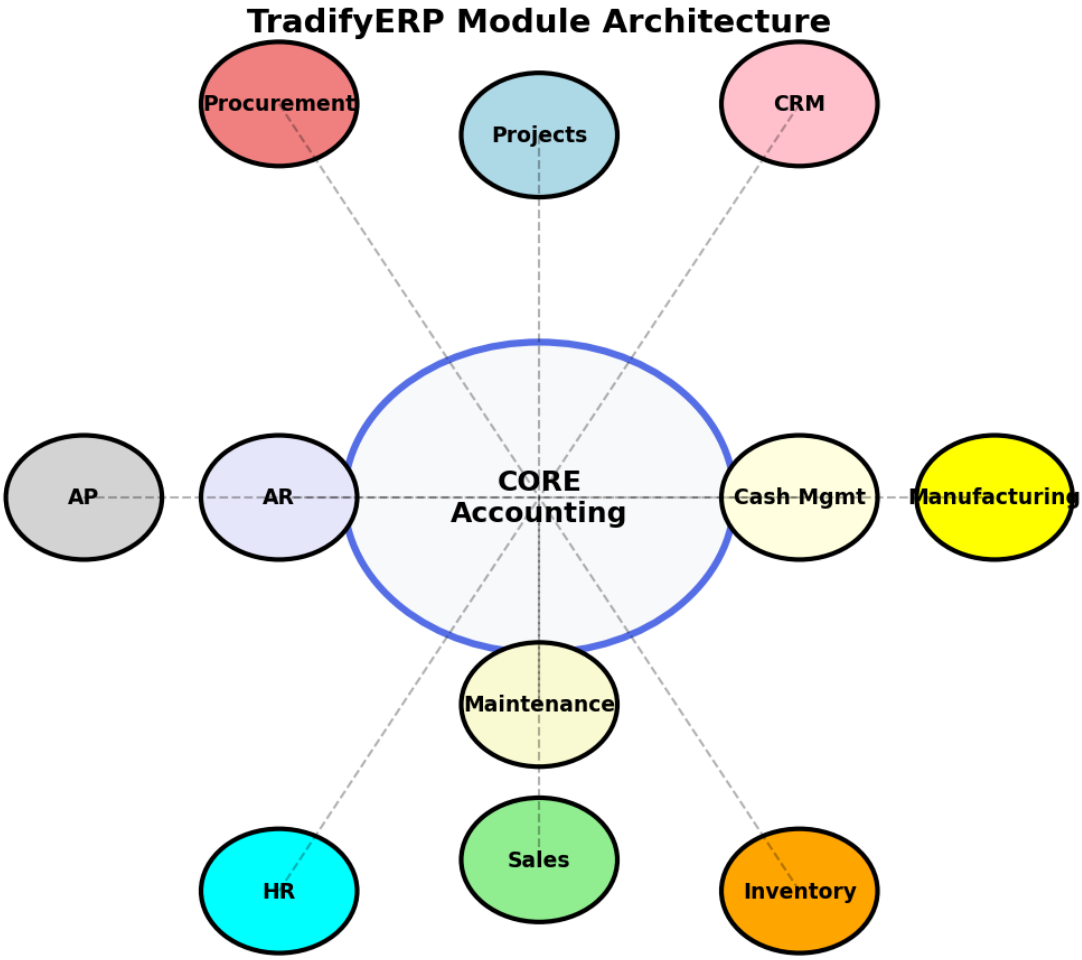
TradifyERP is a comprehensive, enterprise-grade resource planning system designed to manage all aspects of business operations. Built on Django 5.2.8 with Python 3.14, the system comprises 12 fully integrated modules covering financial management, human resources, manufacturing, sales, inventory, procurement, and customer relationship management.

System Statistics

Total Modules	12	Production Ready
Total Models	122+	Database Tables
Total Views	150+	Business Logic
Total Templates	100+	User Interface
Lines of Code	20,000+	Well-Documented
Integrated Workflows	25+	Cross-Module

2. System Architecture

The TradifyERP system follows a modular architecture pattern where each module operates independently while seamlessly integrating with the central accounting core. This design ensures data consistency, enables scalability, and maintains clear separation of concerns.



3. Module Overview

■ Accounting & Finance

Models	29
Views	40
Templates	25

Key Features:

- General Ledger Management
- Trial Balance & Financial Reports
- Fixed Assets & Depreciation
- Accounts Payable/Receivable
- Bank Reconciliation
- Tax Management
- Audit Trail & Period Locks

■ Human Resources

Models	12
Views	15
Templates	12

Key Features:

- Employee Management
- Payroll Processing
- Leave Management
- Attendance Tracking
- Performance Reviews
- Salary History
- Disciplinary Actions

■ Manufacturing

Models	13

Views	18
Templates	15

Key Features:

- Bill of Materials (BOM)
- Work Orders
- Production Planning
- Quality Control
- Routing & Operations
- Resource Management
- Production Reporting

■ Sales Management

Models	3
Views	12
Templates	10

Key Features:

- Sales Orders
- Customer Invoicing
- Revenue Recognition
- GL Integration
- Tax Calculation
- Payment Tracking

■ Inventory Management

Models	8
Views	14
Templates	12

Key Features:

- Multi-warehouse Support
- Stock Movements
- Batch & Serial Tracking
- Stock Adjustments

- Reorder Points
- Inventory Valuation

■ Procurement

Models	6
Views	10
Templates	8

Key Features:

- Purchase Requisitions
- Purchase Orders
- RFQ Management
- 3-way Matching
- Vendor Management
- Procurement Analytics

■ Accounts Payable

Models	4
Views	8
Templates	6

Key Features:

- Vendor Bills
- Payment Processing
- Aging Reports
- Payment Scheduling
- Vendor Statements

■ Accounts Receivable

Models	4
Views	8
Templates	6

Key Features:

- Customer Invoicing

- Payment Collection
- Aging Reports
- Credit Management
- Dunning Process

■ Customer Relationship Management

Models	21
Views	25
Templates	20

Key Features:

- Lead Management
- Opportunity Pipeline
- Quote Management
- Activity Tracking
- Campaign Management
- Loyalty Programs
- Distributor Management

■ Project Management

Models	8
Views	12
Templates	10

Key Features:

- Project Planning
- Task Management
- Time Tracking
- Budget Management
- Resource Allocation
- Milestone Tracking

■ Cash Management

Models	5

Views	8
Templates	6

Key Features:

- Cash Flow Forecasting
- Payment Scheduling
- Treasury Management
- Liquidity Analysis
- Bank Integration

■ Site Maintenance

Models	6
Views	10
Templates	8

Key Features:

- Asset Management
- Work Order Management
- Preventive Maintenance
- Vendor Management
- Cost Tracking

4. Module Interconnections

From Module	To Module	Integration Type	Data Flow
Sales	Accounting	Real-time	Revenue Recognition
Sales	AR	Real-time	Customer Invoices
Manufacturing	Inventory	Real-time	Material Consumption
Manufacturing	Accounting	Batch	Production Costs
Procurement	AP	Real-time	Vendor Bills
Procurement	Inventory	Real-time	Stock Receipts
HR	Accounting	Batch	Payroll Entries
CRM	Sales	Real-time	Opportunities to Orders
Projects	Accounting	Batch	Project Costs
AP	Accounting	Real-time	Payment Entries
AR	Accounting	Real-time	Receipt Entries
Cash Mgmt	Accounting	Real-time	Cash Forecasts